

Hyojoon Park

PH.D. CANDIDATE | RESEARCH SCIENTIST (3D GRAPHICS, VISION, DEEP LEARNING)

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Professional Summary

- My research integrates **Machine Learning** with **Computer Graphics** and **Computer Vision**, building end-to-end systems that combine physics-based simulation, geometry, and image/video processing.
- My work spans **3D reconstruction** of the face, body, hands, and hair, along with **neural rendering**, **XR (VR/AR)**, and **haptics**, with peer-reviewed publications in both Computer Graphics and Medical Imaging.
- I develop **deep learning-based 4D medical image and video compression** using spatiotemporal wavelets, hyperprior-based entropy modeling, latent quantization, and rate-distortion optimization.

Education

University of Wisconsin-Madison

Wisconsin, USA

Ph.D. Candidate in Computer Sciences

Sep. 2021 - Expected Dec. 2026

- **Research area:** Synergistic integration of machine learning in physics-based simulation for computer graphics and medical imaging
- **Advisor:** Professor Eftychios Sifakis, **Co-advisor:** Professor Kevin Elceiri

Seoul National University

Seoul, Korea

M.S. in Mechanical Engineering

Mar. 2017 - Feb. 2019

- **Research area:** Rendering and transparent control of high-performance haptic system
- **Advisor:** Professor Dongjun Lee
- **Thesis:** Dental Simulator with Increased Z-width of Haptic Rendering (also presented at AsiaHaptics 2018)

Technical Contributions

- Extended passive decomposition to SE(3) rigid-body motion with passive midpoint integration for stable, stiff haptic rendering.
- Derived a momentum-based force observer to estimate user-applied forces on multi-DoF haptic devices without force sensors.
- Implemented octree-based voxel rendering for real-time collision detection and material removal in virtual tooth drilling.
- Selected for **Outstanding MS Thesis Presentation Award** [[M.S. thesis presentation](#)]

Technical University of Munich (TUM)

Munich, Germany

B.S. Exchange Student in Mechanical Engineering

Spring 2014

Korea University

Seoul, Korea

B.S. in Mechanical Engineering

Mar. 2010 - Feb. 2017

- Military service: Jun. 2011 - Mar. 2013

Work Experience

Google

Playa Vista, CA, USA

Student Researcher

Sep. 2025 - May 2026

- Developed a real-time, self-supervised neural hair simulator for digital humans, focusing on efficient physics-based animation for XR. Publication-track results in progress.

Technical Contributions

- Designed a GRU-based motion encoder to retain motion history and capture hair inertia, including natural swinging and settling.
- Encoded local and global hair groom structure with a latent transformer, scaling linearly rather than quadratically in strand count.
- Built a StyleGAN2-inspired modulated MLP decoder conditioned on body shape, hairstyle, and motion history for hair animation.
- Trained the neural hair simulator with volumetric elastic, inertial, and collision losses for richer dynamics and faster training.

Nokia Bell Labs

New Providence, NJ, USA

Industrial Metaverse Intern

Jun. 2025 - Aug. 2025

- Developed a real-time 3DGS digital twin with precise physical registration (scale and orientation) and language features for natural-language object queries, as part of "Empowering Digital Twins with Precise Physical Registration and Language-Awareness." Selected as the **1st place winner** among 100+ global interns.
- Follow-up work studies how capture geometry affects 3DGS reconstruction quality. Publication-track results in progress.

Technical Contributions

- Aligned the digital twin with real-world coordinates using ChArUco board correspondences, linear triangulation, nonlinear reprojection-error minimization, and Umeyama similarity alignment.
- Embedded language features in each 3D Gaussian and applied DBSCAN clustering for real-time, text-guided 3D object segmentation.

NVIDIA

Santa Clara, CA, USA

Software Intern - Physics-Based Simulations

May 2024 - Aug. 2024

- Developed a single-image 3D face reconstruction framework trained on synthetic data to generalize from infancy to old age.
- Enabled identity-preserving re-aging and de-aging using explicit age control, differentiable rendering, and adversarial training.

Technical Contributions

- Generated synthetic 3D face meshes from infancy to old age by transferring identity-specific deformations from scanned faces to age-specific face templates.
- Learned the 3DMM identity basis using PCA on skull-aligned face meshes centered on age-specific reference shapes.
- Trained an end-to-end network to predict 3DMM identity and age parameters from a single image, using the pretrained MiVOLO age estimator to supply an age prior.

University of Utah

Salt Lake City, UT, USA

Graduate Research Assistant

Sep. 2019 - Jun. 2021

- Developed and open-sourced a multi-view camera calibration framework (bundle adjustment, ML-based outlier detection) for “Capturing Detailed Deformations of Moving Human Bodies” (SIGGRAPH 2021), tracking 1,000+ uniquely labeled points to reconstruct fine skin deformations of a moving body using a low-cost patterned suit and occlusion-resilient deep networks. [\[code\]](#) [\[paper\]](#)

Technical Contributions

- Built an end-to-end C++ and Python pipeline for multi-camera calibration using checkerboard corner detection and bundle adjustment with Ceres Solver.
- Built a VAE-based outlier detector to filter erroneous checkerboard corners and improve camera calibration.
- Trained a CNN corner detector for a body reconstruction pipeline that rejects invalid image patches, recognizes two-letter codes on the capture suit, and triangulates labeled corners.

Korea Institute of Science and Technology (KIST)

Seoul, Korea

Research Intern

Mar. 2019 - Aug. 2019

- Developed a VR hand-rendering framework for wearable fingertip-tracking devices, reconstructing realistic full-hand poses from fingertip positions alone through motion retargeting and novel IK hinge constraints. Published as “Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices” (IROS 2020). [\[paper\]](#) [\[video\]](#)

Technical Contributions

- Calibrated virtual hand size using fingertip measurements, tracking-device geometry, and SVD-based similarity alignment.
- Retargeted finger joint angles using per-joint scaling and offsets calibrated from open-hand and grasping poses to align rendered and intended hand poses.
- Extended the FABRIK IK solver with planar hinge constraints to prevent unnatural finger bending during real-time hand reconstruction.

Publications

Near-realtime Facial Animation by Deep 3D Simulation Super-Resolution

Hyojoon Park, Sangeetha Grama Srinivasan, Matthew Cong, Doyub Kim, Byungsoo Kim, Jonathan Swartz, Ken Museth, Eftychios Sifakis
ACM Transactions on Graphics (TOG), 2024 (Presented at SIGGRAPH ASIA 2024) [\[paper\]](#) [\[code\]](#) [\[featured video\]](#)

- Upsamples coarse physics-based facial simulations with a neural network for detailed animation at 18 FPS (115x speedup).

Collagen Fiber Centerline Tracking in Fibrotic Tissue via Deep Neural Networks with Variational Autoencoder-based Synthetic Training Data Generation

Hyojoon Park*, Bin Li*, Yuming Liu, Michael S. Nelson, Helen M. Wilson, Eftychios Sifakis, and Kevin W. Eliceiri (*equal contributions)
Medical Image Analysis, 2023 [\[paper\]](#) [\[code\]](#)

- Introduces DuoVAE, pairing a main and auxiliary VAE to generate controllable synthetic data for collagen-fiber centerline tracking.

Capturing Detailed Deformations of Moving Human Bodies

He Chen, **Hyojoon Park**, Kutay Macit, and Ladislav Kavan
SIGGRAPH, 2021 [\[paper\]](#) [\[code\]](#)

- Reconstructs 1,000+ labeled 3D points using CNN corner detection, capture-suit code recognition, and multi-view triangulation.

Adaptive Precision-Enhancing Hand Rendering for Wearable Fingertip Tracking Devices

Hyojoon Park and Jung-Min Park

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020 [\[paper\]](#) [\[video\]](#)

- Uses motion retargeting and FABRIK IK with planar hinge constraints for real-time full-hand reconstruction.

Publications (continued)

Dental Simulator with Increased Z-width of Haptic Rendering

Hyojoon Park, Myungsin Kim, and Dongjun Lee

AsiaHaptics, 2018 [\[paper\]](#) [\[video\]](#)

- Combines passive midpoint integration and passive decomposition to increase stable haptic rendering stiffness by up to 10x.

Rigid-body Collaborative Manipulation among Remote Users with Wearable Cutaneous Haptic Interfaces

Myungsin Kim, WonHa Lee, Hyojoon Park, Junghan Kwon, Yong-Lae Park, and Dongjun Lee

AsiaHaptics, 2018 [\[paper\]](#) [\[video\]](#)

- Enables remote users to jointly manipulate rigid bodies through wearable cutaneous haptic interfaces, using passivity-based control for stable and realistic interactions.

Stretchable Skin-Like Cooling/Heating Device for Reconstruction of Artificial Thermal Sensation in Virtual Reality

Jinwoo Lee, Heayoun Sul, Wonha Lee, Kyung Rok Pyun, Inho Ha, Dongkwan Kim, Hyojoon Park, Hyeonjin Eom, Yeosang Yoon, Jinwook Jung, Dongjun Lee, and Seung Hwan Ko

Advanced Functional Materials, 2020 [\[paper\]](#)

- Reproduces both hot and cold sensations in VR from a single stretchable, skin-like device (230% stretchable), combining heating and cooling in one structure.

Highly Stretchable and Oxidation-resistive Cu Nanowire Heater for Replication of the Feeling of Heat in a Virtual World

Dongkwan Kim, Junhyuk Bang, Wonha Lee, Inho Ha, Jinwoo Lee, Hyeonjin Eom, Myungsin Kim, Jungjae Park, Joonhwa Choi, Jinhyung Kwon, Seungyong Han, Hyojoon Park, Dongjun Lee, and Seung Hwan Ko

Journal of Materials Chemistry A, 2020 [\[paper\]](#)

- Replicates virtual heat on a wearable glove using a 12-pixel stretchable thermal array with laser-patterned copper-nanowire heaters, controlling temperature to within 5% error.

Wearable Cutaneous Haptic Interface with Soft Sensors and IMUs

Yongjun Lee, Myungsin Kim, Yongseok Lee, Hyojoon Park, and Dongjun Lee

Korea Robotics Society Annual Conference, 2018

Design and Performance Evaluation of Wearable Haptic Interfaces

WonHa Lee, Myungsin Kim, Hyojoon Park, and Dongjun Lee

International Conference on Control, Automation and Systems, 2018

Services

SIGGRAPH Asia 2026 Reviewer

2026

Technical Skills

Languages Python, C++, C, CUDA

Frameworks PyTorch, JAX, OpenCV, NumPy/SciPy, OpenGL

Core Areas 3D Vision, Neural Rendering, Physics-based Simulation, Deep Learning, Computer Graphics, 3D Reconstruction, Data Compression

Tooling Linux, Git, Docker, CMake, $\text{\LaTeX}$

Teaching Assistant

Computer Graphics (CS559) *University of Wisconsin-Madison*

Spring 2022

Computer Graphics (CS559) *University of Wisconsin-Madison*

Fall 2021

Interactive Computer Graphics (CS6610) *University of Utah*

Spring 2021

Computer Graphics (CS4600) *University of Utah*

Fall 2020

System Analysis in Mechanical Engineering *Seoul National University*

Spring 2018